

SOFIYA KITTUR

Robotics & Automation Engineer

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PROFILE

Robotics & Automation engineering student with hands-on experience in computer vision (object detection, tracking, counting), embedded systems (Arduino, RPi, ESP32), and automation projects. Coursework and projects in ROS2, Gazebo, and Python for robotics. Seeking opportunities to learn and contribute to real-world robotics applications, especially in healthcare and safety-critical environments.

TECHNICAL SKILLS

- Robotics: Experience with ROS2 and Gazebo through coursework and projects
- Computer Vision: Real-time object detection, tracking, counting pipelines (99%+ mAP)
- Programming: Python, C++, web frontend (HTML)
- Embedded Systems: Arduino, ESP32, Raspberry Pi, sensor integration, IoT device development
- Relevant Coursework: Control Systems, Robot Mechanics, Embedded System Design, AI

WORK EXPERIENCE

Ravtor Mobility Pvt. Ltd. | Automation and Tech Lead

Dec 2025 – Mar 2026

- Led R&D on EV bike architecture, battery systems, and motor technologies, informing design decisions for vehicle prototypes
- Engineered a patent-pending electromechanical smart lock from initial concept through working prototype
- Designed and integrated a GPS-based tracking module for real-time vehicle location monitoring
- Developed a mobile app interface for IoT-based lock control, enabling remote lock/unlock and device connectivity

Synart Automation Pvt. Ltd. | Intern

Jul 2025 – Sep 2025

- Designed a drone PCB end-to-end, including component selection and board layout
- Contributed to small-scale electronics projects and supported testing/prototyping cycles across the product pipeline

PROJECTS

Electromechanical Smart Lock (College Project - Selected for E CO Op)

2026

- Designed and built a smart lock system integrating mechanical design (disc brake locking mechanism), electrical components (LED status indicators, alarm), and a mobile app for remote lock control
- Developed the mobile app to connect with the lock via Bluetooth, allowing users to lock/unlock and receive alerts for disturbances
- Implemented safety features, including an alarm that sounds if the lock is disturbed, prompting the owner to re-adjust the vehicle for a secure fit
- Project was selected by college's E CO Op program for further development into a marketable product or patent filing

Aequs Ltd. - Vision-Based Multi-SKU Production Monitoring | Computer Vision Engineer (Industry-Collaborated College Project)

2026

- Built a single-camera computer vision system that converts products moving on a conveyor into real-time, SKU-wise production data for high-volume manufacturing
- Led customer discovery (20 stakeholder interviews across 4 segments) that refined the initial hypothesis into a validated, high-volume conveyor-manufacturing use case
- Designed the detect → classify → track → count → monitor pipeline (ByteTrack for tracking, virtual line-crossing for counting); trained on a custom 12,749-image, 7-SKU dataset achieving 99.56% precision, 99.57% recall, 99.49% mAP@50
- Benchmarked against manual counting, PLC/cycle counters, and commercial vision systems (Keyence, Cognex, SICK); authored a TRL 4 → TRL 6 validation roadmap covering counting accuracy, robustness, and endurance testing

Hosmani Neutraceuticals Pvt. Ltd. - Automated Water Level Control & Monitoring System | Automation Engineer (Industry-Collaborated College Project)

2025

- Partnered with Hosmani Neutraceuticals on an automated water-level control and monitoring system for algae farm ponds, as part of a multi-disciplinary engineering team
- Designed the valve actuation mechanism for automated water-level regulation, integrating salinity and temperature sensors
- Built the HMI layout for operator interaction, system diagnostics, and real-time data visualization

Temperature Monitoring System (Academic Project) | Embedded System Developer

- Designed a microcontroller-based temperature monitoring system with cloud-based data storage for real-time and historical analysis, under the guidance of the founder of Saureno Energrene Pvt. Ltd.
- Contributed to a Mechanical Engineering Department initiative building integrated monitoring solutions for industrial applications

3D Modeling for University Projects | Project Member

- Developed detailed 3D models for university engineering projects using SolidWorks and AutoCAD

LEADERSHIP & ENTREPRENEURSHIP

Ten4teen | Founder

Mar 2026 – Present

- Founded and launched a platform connecting startups with engineering students for real-world project opportunities, going live publicly on July 1, 2026; secured 25+ startup partnerships in Hubli via the Deshpande Startups ecosystem and built a 100+ student pre-launch waitlist

- Own end-to-end product development, outreach, and partnership strategy as sole founder

EDUCATION

KLE Technological University, BVB, Hubli, Karnataka | *B.Tech, Automation & Robotics Engineering* Expected Sep 2027